Lahore AQI Predictor

Air Quality Index Forecasting for Lahore, Pakistan

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3-Day AQI Forecast Using Machine Learning

End-to-End ML Pipeline with Automated Data Collection,

Feature Engineering, Model Training, and Real-Time Predictions

1. Executive Summary

This report documents the Lahore AQI Predictor, an end-to-end machine learning system for forecasting Air Quality Index (AQI) in Lahore, Pakistan for the next 3 days (72 hours). The system uses a 100% serverless architecture with automated data collection, feature engineering, model training, and real-time predictions through an interactive web dashboard.

Lahore is one of the most polluted cities globally, particularly during the smog season (October-February) when AQI values regularly exceed 300. Accurate AQI forecasting is critical for public health advisories and planning.

2. System Architecture

The system consists of four main pipelines:
1. Feature Pipeline: Fetches raw weather and pollutant data from AQICN and Open-Meteo APIs, computes 35 features, and stores them in the Supabase Feature Store.
2. Training Pipeline: Fetches historical features from Supabase, trains multiple ML models, evaluates performance, and registers the best model in Hopsworks Model Registry.
3. Inference Pipeline: Loads the model, generates 72-hour recursive forecasts conditioned on future weather, and enforces an automatic model fallback cascade.
4. Web Dashboard: Displays real-time predictions, daily forecast cards, historical trends, and SHAP model explanations via Streamlit + Flask.

2.1 Technology Stack

Technology	Purpose
Python 3.12	Core programming language
Scikit-learn	Ridge Regression, Random Forest, metrics & scalers
XGBoost	High-performance gradient boosting time-series forecaster
Supabase (PostgreSQL)	Serverless Feature Store with instant HTTP ingestion
Hopsworks	Serverless Model Registry for versioning & artifacts
AQICN + Open-Meteo	Real-time telemetry and historical backfill APIs
Streamlit + Flask	Full-width glassmorphic UI & RESTful API
GitHub Actions	Automated hourly/daily serverless CI/CD
SHAP	Model explainability & feature attribution

2.2 Feature Store Architecture: Why Supabase Over Hopsworks

While Hopsworks was evaluated for feature storage, several critical operational limitations led to adopting Supabase PostgreSQL as the primary serverless feature store:

1. Free Tier Limits & Paywalls: Hopsworks Cloud enforces strict storage and compute credit limits on free accounts. Continuous hourly feature ingestion rapidly exhausts free credits, triggering paywall blocks or project suspensions.
2. Inactivity Shutdowns & Network Latency: Inactive Hopsworks free clusters hibernate, causing 504 Gateway Timeouts, 30-60 second cold-start delays, and failed CI/CD workflow executions.
3. Heavy Dependency Overhead: Hopsworks HSFS relies on PySpark and JVM dependencies, which frequently cause installation failures and memory limits on lightweight serverless containers.
4. Supabase Advantages: Supabase provides a 100% free serverless PostgreSQL database with generous limits, sub-100ms PostgREST HTTP queries, atomic primary key upserts for duplicate prevention, and zero JVM requirements.
5. Optimal Separation of Concerns: Hopsworks is retained for its core strength as a Model Registry, while Supabase

handles high-frequency, reliable feature storage.

2.3 Model Architecture: Why XGBoost Replaced TensorFlow

TensorFlow/Keras LSTM networks were benchmarked for sequence modeling. However, TensorFlow introduces a ~1.5 GB binary footprint, slow CPU training on CI/CD runners, and frequent platform incompatibilities. XGBoost and Scikit-learn Random Forest delivered superior R2 scores, faster convergence, and sub-second inference with zero heavy runtime dependencies.

3. Data Description

The system collects hourly data from multiple sources. Weather data includes temperature, humidity, wind speed/direction, pressure, and precipitation from Open-Meteo. Air quality data includes PM2.5, PM10, NO2, SO2, O3, and CO concentrations, plus the US AQI index.

3.1 Feature Categories

Category	Count	Description
Weather	8	Temperature, humidity, wind, pressure, cloud cover
Pollutants	6	PM2.5, PM10, NO2, SO2, O3, CO
Time-based	6	Hour, day, month, weekend, season
Lahore-specific	5	Smog season, crop burning, brick kilns, wind direction
Derived/Lag	10	Rolling stats, lag features, interaction terms
TOTAL	35	All model input features

4. Model Performance Comparison

Three ML models were trained and evaluated using time-based train/validation/test splits (80/10/10) to prevent temporal data leakage. Models were evaluated using RMSE (Root Mean Squared Error), MAE (Mean Absolute Error), R-squared, and MAPE metrics.

Model	RMSE	MAE	R2	MAPE
RIDGE	0.20	0.08	1.0000	0.05%
RANDOM_FOREST	14.11	2.96	0.9262	1.29%
XGBOOST	14.94	2.91	0.9172	1.20%

4.1 Model Fallback Chain

The system implements a model fallback chain for 100% production reliability:
Primary: XGBoost -> Fallback 1: Random Forest -> Fallback 2: Ridge -> Emergency: Last known AQI values.

Each model undergoes validation checks before serving predictions. If it produces errors, NaN values, negative AQI, or values exceeding 600, the system automatically cascades to the next model.

5. AQI Categories & Health Advisories

AQI Range	Category	Color Code
0-50	Good	#00e400
51-100	Moderate	#ffff00
101-150	Unhealthy for Sensitive Groups	#ff7e00
151-200	Unhealthy	#ff0000
201-300	Very Unhealthy	#8f3f97
301-500	Hazardous	#7e0023

6. Lahore-Specific Analysis

Lahore has unique pollution patterns driven by several factors:

- Smog Season (Oct-Feb): Temperature inversions trap pollutants near the surface, causing severe air quality deterioration. AQI regularly exceeds 300.
- Crop Burning (Oct-Nov): Stubble burning in Punjab contributes significantly to PM2.5 levels during the post-harvest season.
- Brick Kilns (Nov-Mar): Traditional brick kilns operate during winter months, adding to industrial emissions.
- Cross-border Pollution: Easterly winds carry pollution across the border.
- Rain Effect: Rainfall is the primary natural air cleanser in Lahore.

7. Automation & CI/CD

The system uses GitHub Actions for automated pipeline execution:

- Feature Pipeline: Runs every hour to fetch fresh weather and air quality data, compute features, and store them in the Feature Store.
- Training Pipeline: Runs daily at 2 AM UTC (7 AM PKT) to retrain models with the latest data and update the Model Registry.

This ensures the prediction models are always up-to-date with the latest data patterns and seasonal changes.

8. Conclusion & Deliverables

The Lahore AQI Predictor provides a comprehensive, automated solution for air quality forecasting in Lahore. Key achievements:

1. End-to-end automated pipeline from data collection to 72-hour prediction serving.
2. 100% Serverless Architecture: Supabase PostgreSQL Feature Store + Hopsworks Model Registry.
3. Production ML Suite (XGBoost, Random Forest, Ridge) with automated fallback cascade.
4. Lahore-specific atmospheric features capturing winter smog, temperature inversion, and wind dispersion.
5. Interactive full-width dashboard with real-time forecasts, WHO health advisories, and daily cards.
6. SHAP-based model explainability for transparent, interpretable predictions.
7. Fully automated CI/CD using GitHub Actions for hourly feature ingestion and daily retraining.